

Numpy * Python

Properties of numpy

① Shape in numpy.

```
import numpy as np  
arr = np.array([[1,2,3],  
                [3,4,5]])
```

```
print(arr.shape)
```

② Size → to number of element

```
print(arr.size)
```

③ ndim → number of dimensions

```
print(arr.ndim)
```

④ dtype → data type of element nature.

```
print(arr.dtype)
```

⑤ astype → to change the data type

```
arr.astype(new datatype)  
arr.astype(int)
```

⑥ function (What it does.)

np.sum(array) → add all

np.mean(array) → find average

np.min(array) → find minimum

np.max(array) → find maximum

np.std(array) → find ~~stare~~ standard deviation.

np.var(array) → find variance.

Indexing and slicing.

① access in indexing

```
import numpy as np
arr = np.array([10, 20, 30, 40])
print(arr[0])
print(arr[2])
print(arr[-1])
```

② access in slicing

```
arr[start; ending; step]
arr[2:4:1]
```

③ fancy indexing: selecting multiple element at once

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50, 60])
print(arr[0, 3, 4, 5])
```

④ Boolean masking:

```
print(arr[arr > 25])
```

Reshaping and manipulating

reshape (row, column) // if dimension match

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8])
reshaped_arr = arr.reshape(2, 4)
print(reshaped_arr)
```

// does not create a copy it change the original data

① flatten :- It convert multi dimension to 1 D.

(i) `.ravel()` → views

(ii) `.flatten()` → copy.

② insert :- `.insert(arr, index, value, axis)`

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4, 5])
```

```
new_arr = np.insert(arr, 3, 10, axis=0)
```

```
print(new_arr)
```

③ append :- add at last

```
np.append([4, 5, 6])
```

④ Concatenate :- add to arr

⑤ delete :- `np.delete(array, index, axis=None)`

⑥ Stacking :- Combine array

(i) `vstack()` :- Vertical

(ii) `hstack()` :- Horizontal

⑦ splitting

(i) `np.split(arr, 2)`

(ii) `np.hsplit()`

(iii) `np.vsplit()`

Broadcasting in numpy.

① Discount

```
import numpy as np
arr price = np.array([100, 200, 300, 400])
discount = 10
final_price = price - (price * discount / 100)
print(final_price)
```

Broadcasting has three rule.

- ① matching dimensions
- ② expanding single element $[1, 2, 3] + 10$
- ③ Incompatible shape gives error.

Vectorization

```
import numpy as np
arr1 = np.array([3, 5, 7])
arr2 = np.array([1, 2, 6])
result = arr1 + arr2
print(result)
```

① isna - Find missing number.

```
import numpy as np
arr = np.array([1, 2, np.nan, 4, 6])
print(np.isnan(arr))
```

② handling missing value with number.

```
arr = np.nan_to_num(arr, nan=100)
```

③

infinte

print(np.isinf(arr))